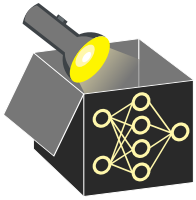
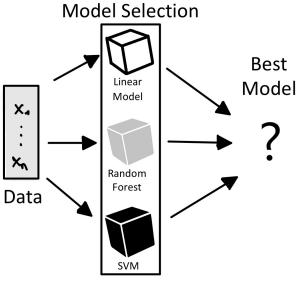
# Interpretable Machine Learning



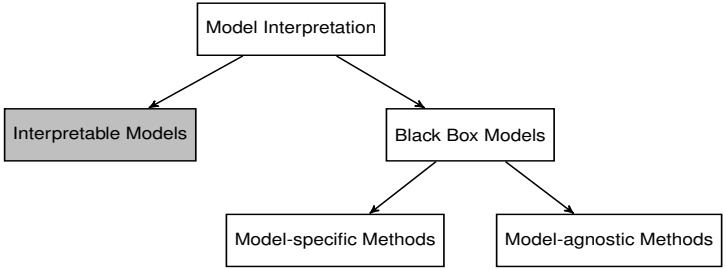
## Dimensions of Interpretability

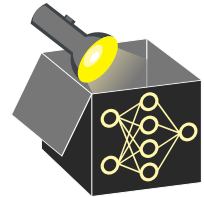



**Learning goals**

- Intrinsic vs. model-agnostic methods
- Different types of explanations
- Local vs. global methods
- Model or learner explanations – with or without refits
- Levels of interpretability

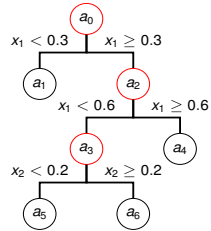
# INTRINSIC VS. MODEL-AGNOSTIC



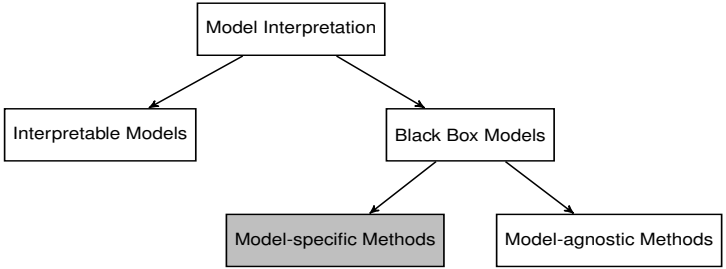



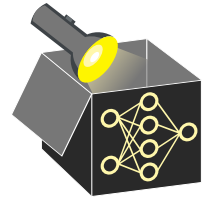
Intrinsically interpretable models:

- Examples: linear model, decision tree, decision rule, GLMs
- Interpretable because of simple model structure, e.g., weighted combination of feature values or tree structure
- Difficult to interpret with many features / complex interactions



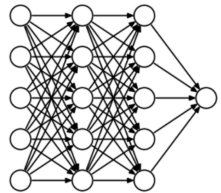
# INTRINSIC VS. MODEL-AGNOSTIC



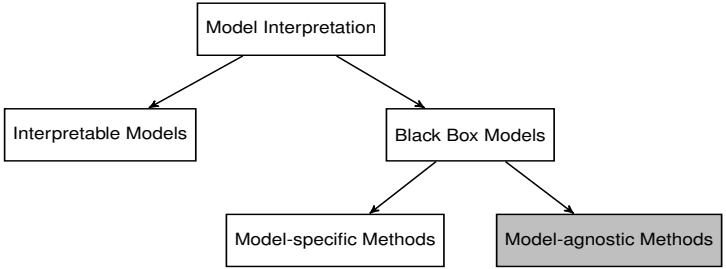



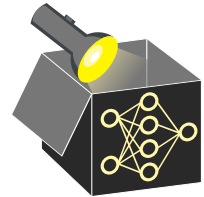
Model-specific methods:

- Interpretation method applicable to a specific ML model
- Example: implicitly integrated feature interpretation methods in tree based models, e.g., Gini Importance
- Advantage: Can exploit model structure
- Visualize activations of NNs



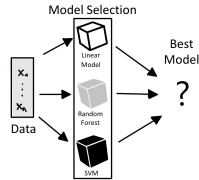
# INTRINSIC VS. MODEL-AGNOSTIC



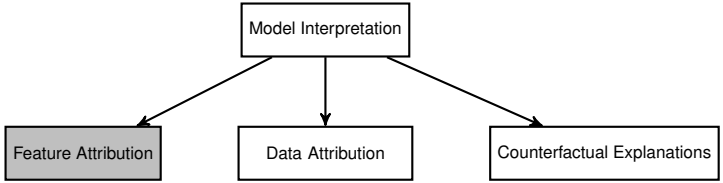



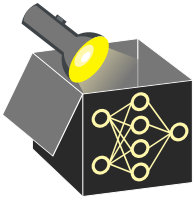
Model-agnostic methods:

- In ML: Tune over many model classes
  $\rightsquigarrow$ Unknown which model is best / deployed
  $\rightsquigarrow$ Need for interpretation methods applicable to any model
- Applied after training (post-hoc)
- Applicable to intrinsically interpretable models
  $\rightsquigarrow$ provides insights into other types of explanations



# TYPES OF EXPLANATIONS

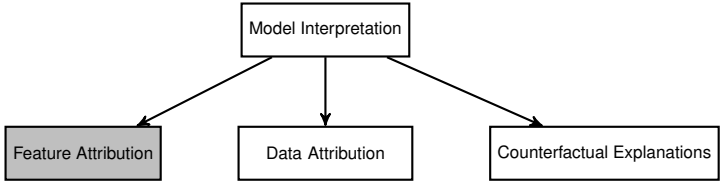


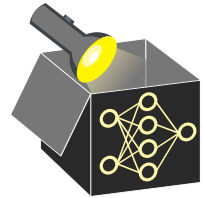



Feature Attribution:

- Produce explanations on a per-feature level, e.g., feature effects or feature importance
- Vary feature values, inspect change of model prediction, model variance or model error

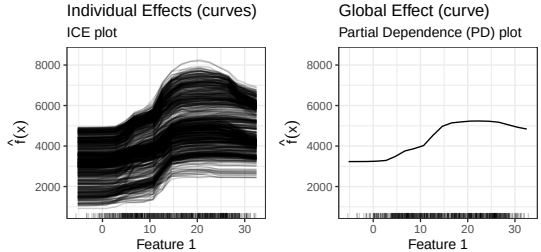
# TYPES OF EXPLANATIONS



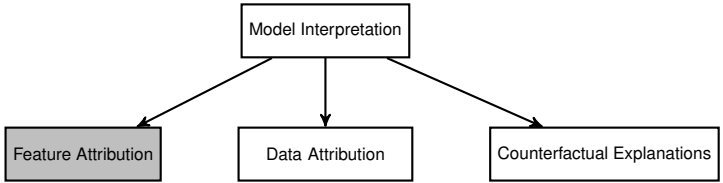



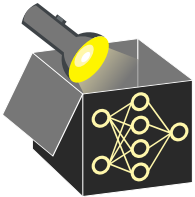
**Feature Effects** indicate the change in prediction due to changes in feature values.

- Model-agnostic methods: ICE curves, PD plots . . .
- Pendant in linear models: Regression coefficient $\theta_j$
- Further examples: Saliency Maps, model-agnostic methods such as SHAP and LIME



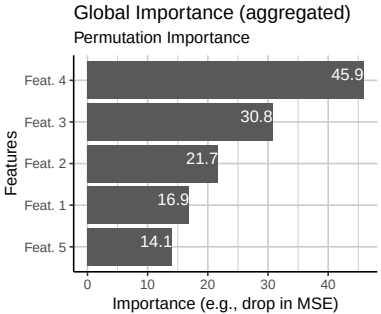
# TYPES OF EXPLANATIONS



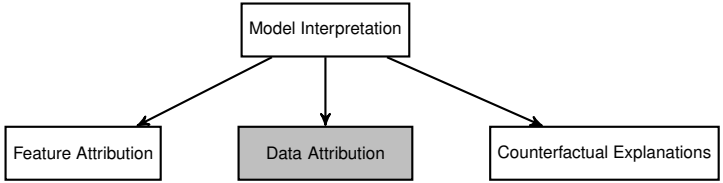



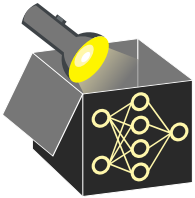
**Feature importance** methods rank features by how much they contribute to the predictive performance or prediction variance of the model.

- Model-agnostic methods: PFI, . . .
- Pendant in linear models: t-statistic, p-value (significant effect)



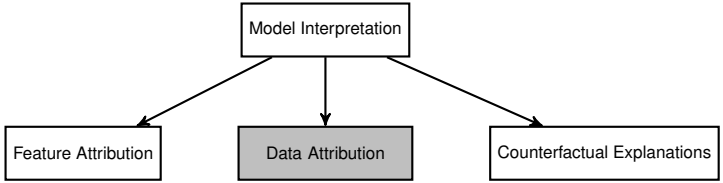
# TYPES OF EXPLANATIONS

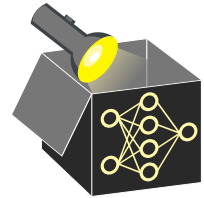


Data Attribution: Identify training instances most responsible for a decision (e.g. Influence Functions)

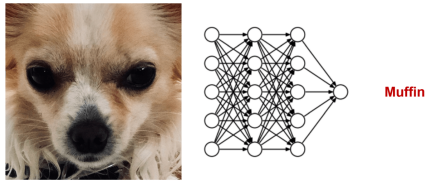
# TYPES OF EXPLANATIONS



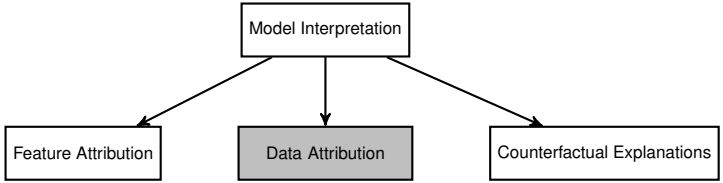
Data Attribution: Identify training instances most responsible for a decision (e.g. Influence Functions)

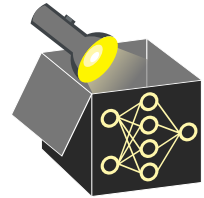
**Example:** Consider a model which should distinguish muffins and dogs




How does this incorrect prediction come about?

# TYPES OF EXPLANATIONS



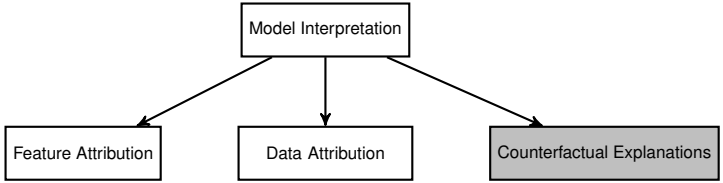



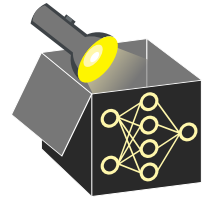
Data Attribution: Identify training instances most responsible for a decision (e.g. Influence Functions)
Look at training data: Which data points caused the model prediction?

Method searches for the most similar images and bases the decision on them

- ⇝ Training images looking most like new input show a muffin
- ⇝ Wrong output (muffin instead of dog)

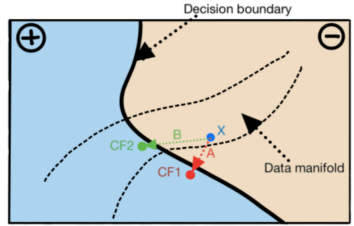
# TYPES OF EXPLANATIONS



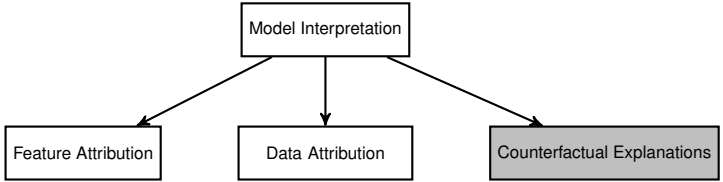



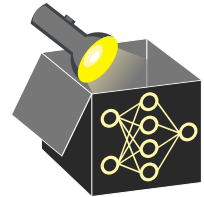
Counterfactual Explanations:

- Identify smallest necessary change in feature values so that a desired outcome is predicted
- Contrastive explanations
- Diverse counterfactuals
- Feasible & actionable explanations

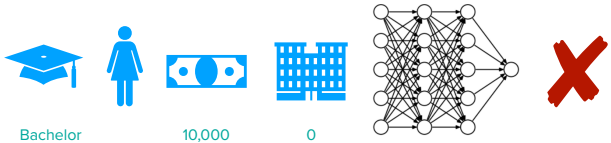


# TYPES OF EXPLANATIONS

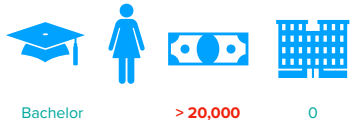


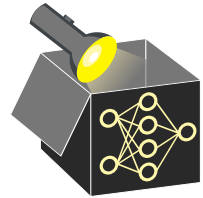
**Example** (loan application):




What can a person do to obtain a favorable prediction from a given model ?

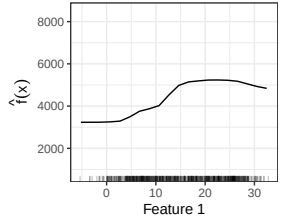


# GLOBAL VS. LOCAL



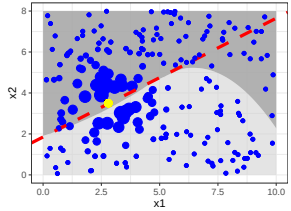
Global interpretation methods explain the model behavior for the entire input space by considering all available observations:

- Permutation Feature Importance (PFI)
- Partial Dependence (PD) plots
- Accumulated Local Effect (ALE) plots
- ...

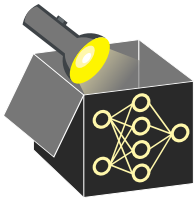



Local interpretation methods explain the model behavior for single data instances:

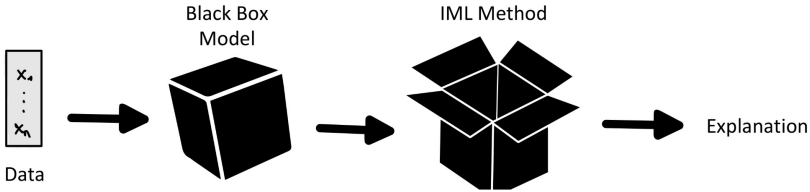
- Individual Conditional Expectation (ICE) curves
- Local Interpretable Model-Agnostic Explanations (LIME)
- Shapley values, SHAP
- ...



# FIXED MODEL VS. REFITS

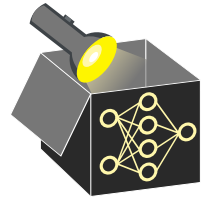


- Input of global interpretation methods: model + data, output: explanations
  $\rightsquigarrow$ Explanations can be viewed as statistical estimators

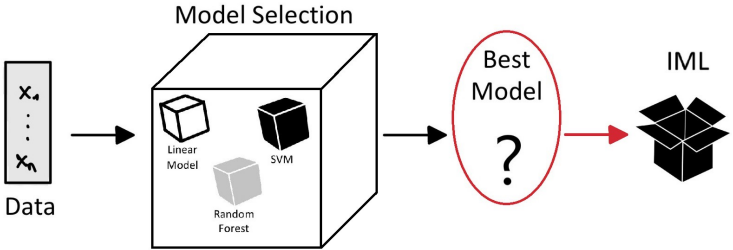



- Situation in ML: Deployed model is trained on all available data
  $\rightsquigarrow$ No unseen test data left to, e.g., reliably estimate performance
  $\rightsquigarrow$ IML method could use same data model was trained on
  $\rightsquigarrow$ But: Some IML methods rely on measuring loss requiring unseen test data
- Alternative: Explain the inducer that created the model (instead of a fixed model)
  $\rightsquigarrow$ Idea: Use resample strategies (e.g., 4-fold CV) as in performance estimation
  $\rightsquigarrow$ Requires refitting

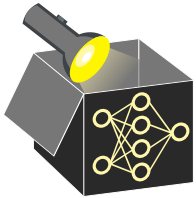
# LEVELS OF INTERPRETABILITY



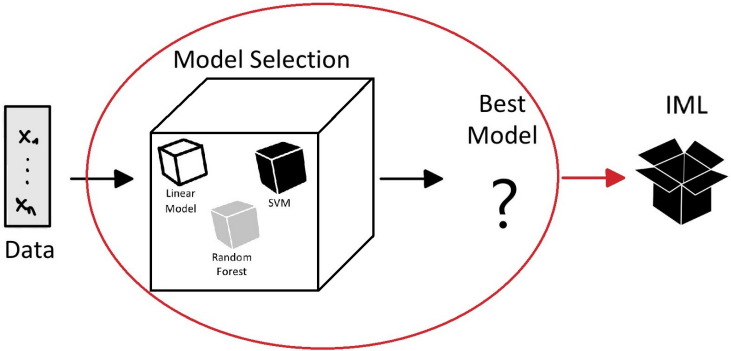
| | Research Question | Objects of analysis |
|---|---|---|
| 1st level view | How to explain a given model fitted on a data set? | (deployed) model $\theta \mapsto \hat{f}(\theta)$ |



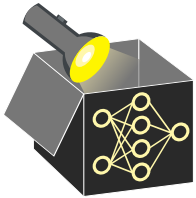
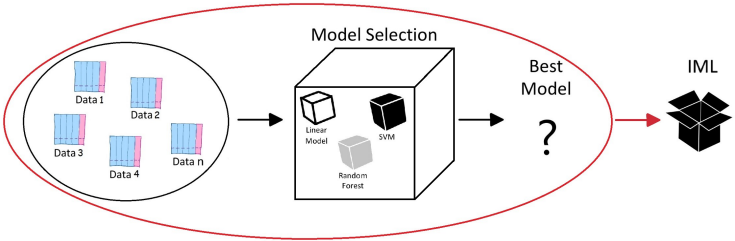
# LEVELS OF INTERPRETABILITY



| | Research Question | Objects of analysis |
|---|---|---|
| 1st level view | How to explain a given model fitted on a data set? | (deployed) model $\theta \mapsto \hat{f}(\theta)$ |
| 2nd level view | How does an optimizer choose a model based on a data set? | Model selection process (e.g., decisions made by AutoML systems or HPO process) |



# LEVELS OF INTERPRETABILITY



| | Research Question | Objects of analysis |
|---|---|---|
| 1st level view | How to explain a given model fitted on a data set? | (deployed) model $\theta \mapsto \hat{f}(\theta)$ |
| 2nd level view | How does an optimizer choose a model based on a data set? | Model selection process (e.g., decisions made by AutoML systems or HPO process) |
| 3rd level view | How do data properties relate to performance of a learner and its hyperparameters? | properties of ML algorithms in general (benchmark) |